

Unknotting number equals genus for the Baker–Kegel and Himeno L -space knots

2026-10-10

Abstract

For an L -space knot $u \geq g_4 = g$. We show $u = g$ for two infinite families of hyperbolic L -space knots that are not braid positive, or not known to be: the knots K_k of Baker and Kegel and the knots K_n of Himeno. Both are members of a two-parameter family $H(n, m)$ of closed $2n$ -braids, and $u(H(n, m)) = g_4(H(n, m)) = (mn^2 - n + 2)/2$ by an explicit unknotting. The same argument settles the L -space $(2, q)$ -cables of $T(2, 2k + 1)$ that are not positive braids. The L -space hypothesis is not superfluous: the mirror of $12n_{642}$ is fibered and strongly quasipositive, with $g = 2$ and $u \geq 3$.

1 Context

An L -space knot is fibered and strongly quasipositive, and $g = g_4 = \tau$ [11, 9, 4]. Since $u \geq g_4$ for every knot, $u \geq g$, and the statement $u = g$ for every L -space knot is the statement $u \leq g$. It was produced as candidate K32 by a program that fits inequalities to the KnotInfo table [7]; the companion note on $\det \leq 1 + |\sigma|$ describes the program. We did not find the statement in the literature.

It holds for braid positive knots: $u \leq g$ is due to Rudolph [12], and Kegel, Lewark, Manikandan, Misev, Mousseau and Silvero show that every positive braid diagram of a genus g knot has g crossings whose change gives the unknot [6]. This covers the torus knots, the $(1, 1)$ L -space knots, which are closures of positive braids [10, Theorem 2], and all L -space knots of the SnapPy census but one [1]. It also holds for strongly quasipositive 3-braid knots [8]. What remains are the L -space knots that are not braid positive. The ones we know are the families of Baker–Kegel [1] and Himeno [5] and some iterated torus knots, such as the $(2, 3)$ -cable of the trefoil [1, Example 1].

2 The family

Fix $n \geq 2$. In the braid group B_{2n} let

$$L_k = \sigma_{n-k} \sigma_{n-k+2} \cdots \sigma_{n+k} \quad (0 \leq k \leq n-1), \quad X_n = L_0 L_1 \cdots L_{n-1} L_{n-2} \cdots L_0,$$

$$T_n = \sigma_1^{-1} \cdots \sigma_{n-1}^{-1} \sigma_n \sigma_{n-1} \cdots \sigma_1 \sigma_1 \sigma_2 \cdots \sigma_n,$$

and for odd $m \geq 1$ let $H(n, m)$ be the closure of $X_n^m T_n$. The letters of each L_k commute. X_n is the n -cable of a positive crossing between two bundles of n strands. The closure is a knot: the permutation of X_n^m exchanges i and $n + i$, that of T_n is an n -cycle on $1, \dots, n$, and their product is a $2n$ -cycle.

Since $X_2 = \sigma_2 \sigma_1 \sigma_3 \sigma_2$ and $T_2 = \sigma_1^{-1} \sigma_2 \sigma_1 \sigma_1 \sigma_2$, the knot $H(2, 2k + 1)$ is the knot K_k of Baker and Kegel [1], a hyperbolic L -space knot of genus $4k + 2$ for $k \geq 1$, not braid positive for $k = 1$ (K_1 is the

census knot o9_30634) and not known to be braid positive for $k \geq 2$. The knot $H(n, 3)$ is the knot K_n of Himeno [5], hyperbolic for $n \geq 2$, and for even n an L -space knot of genus $(3n^2 - n)/2 + 1$ that is not braid positive.

3 The theorem

Theorem 1. *For $n \geq 2$ and odd $m \geq 1$, $u(H(n, m)) = g_4(H(n, m)) = (mn^2 - n + 2)/2$.*

Corollary 1. *$u = g$ for every Baker–Kegel knot K_k and every Himeno knot K_n that is an L -space knot: all K_k [1], all K_n with n even [5], and K_3 and K_5 , which knot Floer homology shows to be L -space knots.*

Proof of the corollary. For an L -space knot $g = g_4$. □

Proof of Theorem 1. Lower bound. The word $X_n^m T_n$ has $mn^2 + 2n$ positive and $n - 1$ negative letters, so write $w = mn^2 + n + 1$, on $2n$ strands. Rudolph’s slice–Bennequin inequality [13] gives $g_4 \geq (w - 2n + 1)/2 = (mn^2 - n + 2)/2$, and $u \geq g_4$.

Upper bound. A crossing change in the closed braid replaces a letter $\sigma_i^{\pm 1}$ by $\sigma_i^{\mp 1}$. We make $(mn^2 - n + 2)/2$ of them and reach the unknot.

Step 1. Read backwards, X_n is $L_0 \cdots L_{n-2} L_{n-1} L_{n-2} \cdots L_0$ with the letters of each layer permuted, so it represents the same braid; hence the word obtained from X_n by inverting every letter represents X_n^{-1} . Changing all crossings of the second, fourth, $\dots$, $(m-1)$ -st copy of X_n turns $X_n^m T_n$ into $X_n T_n$, with $n^2(m-1)/2$ changes.

Step 2. Write $X_n = CL_{n-1}\bar{C}$ with $C = L_0 \cdots L_{n-2}$ and $\bar{C} = L_{n-2} \cdots L_0$. Changing the $n(n-1)/2$ crossings of $\bar{C}$ replaces it by C^{-1} .

Step 3. $T_n = a\sigma_1 \cdots \sigma_n$ with $a = (\sigma_{n-1} \cdots \sigma_1)^{-1} \sigma_n (\sigma_{n-1} \cdots \sigma_1)$. Changing the crossing σ_n inside a replaces a by a^{-1} . In all, $n^2(m-1)/2 + n(n-1)/2 + 1 = (mn^2 - n + 2)/2$ changes, giving the braid $CL_{n-1}C^{-1}R$ with $R = a^{-1}\sigma_1 \cdots \sigma_n \in B_{n+1}$.

Step 4: the closure of $CL_{n-1}C^{-1}R$ is the unknot. It is the closure of the conjugate $L_{n-1} \cdot C^{-1}RC$. Call the strands at the top of X_n $A_1, \dots, A_n$ (positions $1, \dots, n$) and $B_1, \dots, B_n$ (positions $n+1, \dots, 2n$). Every crossing of X_n is between an A strand and a B strand, with the A strand on the same side, say over, every time. C performs the crossings of A_i with B_j for $j < i$, after which A_i is at position $2i-1$ and B_i at $2i$, and L_{n-1} crosses A_i with B_i . In the conjugate, after L_{n-1} the strand B_i is at position $2i-1$ and A_i at $2i$. The word C^{-1} moves the strands at odd positions back to positions $1, \dots, n$ and those at even positions to $n+1, \dots, 2n$; since C^{-1} undoes C , the strand coming from an even position passes under every strand it meets, in C^{-1} and again in C . So R acts on $B_1, \dots, B_n, A_1$ and leaves $A_2, \dots, A_n$ alone, and each A_i with $i \geq 2$, followed from its crossing in L_{n-1} once around the closed braid back to that crossing, lies under every strand it meets. Pushing that arc below the diagram and shrinking it removes the strand and its crossing. After doing so for $i = n, \dots, 2$ the closure is that of

$$\sigma_1 \cdot (\sigma_2^{-1} \cdots \sigma_n^{-1}) \cdot R \cdot (\sigma_n \cdots \sigma_2) \in B_{n+1},$$

where σ_1 is the crossing of A_1 with B_1 , the factor before R moves A_1 to position $n+1$ under $B_2, \dots, B_n$, and the factor after R moves the strand then in position $n+1$ back in the same way. Conjugating by $\sigma_n \cdots \sigma_2$ and using $\sigma_n \cdots \sigma_2 \sigma_1 \sigma_2^{-1} \cdots \sigma_n^{-1} = (\sigma_{n-1} \cdots \sigma_1)^{-1} \sigma_n (\sigma_{n-1} \cdots \sigma_1) = a$, which follows by induction from $\sigma_{i+1} \sigma_i \sigma_{i+1}^{-1} = \sigma_i^{-1} \sigma_{i+1} \sigma_i$, the closure is that of $aR = \sigma_1 \sigma_2 \cdots \sigma_n \in B_{n+1}$, the unknot. □

Without the change in Step 3 the same reduction gives the closure of $a^2 \sigma_1 \cdots \sigma_n$, a knot of genus one (its braided surface has $n+2$ bands on $n+1$ disks); the change in Step 3 unknots it.

4 Cables

Corollary 2. *Every L -space $(2, q)$ -cable of $T(2, 2k + 1)$ has $u = g$.*

Proof. The cable is the closure of $X_2^{2k+1}\sigma_1^e$ with $e = q - 2(2k + 1)$, and it is an L -space knot exactly when $q \geq 2(2k - 1)$ [2, 3], that is $e \in \{-3, -1, 1, 3, \dots\}$. For $e \geq 1$ the word is positive. For $e = -1$, Step 1 with k cancelled copies of X_2 ($4k$ changes) leaves $X_2\sigma_1^{-1}$, whose closure is the unknot, and $g = 4k$. For $e = -3$, cancelling $k - 1$ copies ($4k - 4$ changes) leaves $X_2^3\sigma_1^{-3}$, the $(2, 3)$ -cable of the trefoil, which three changes unknot (its first three letters); $g = 4k - 1$. $\square$

5 The hypothesis is needed

Proposition 1. *The mirror of $12n_{642}$ is fibered and strongly quasipositive, has genus 2, is not an L -space knot, and has unknotting number at least 3.*

Proof. Knot Floer homology gives genus 2, fiberedness and $|\tau| = 2$; the chirality with $\tau = g$ is strongly quasipositive, since a fibered knot with $\tau = g$ is [4]. Its total rank is 37, more than $2g + 1$, so it is not an L -space knot. A Seifert matrix V has $\det(V + V^T) = \pm 27$ and $V + V^T$ has nullity 3 over $\mathbb{F}_3$, so the first homology of the double branched cover is $(\mathbb{Z}/3)^3$ and needs three generators; by Wendt [14] this is a lower bound for the unknotting number. $\square$

KnotInfo lists $u(12n_{642}) \in \{3, 4\}$. So $u = g$ fails for fibered strongly quasipositive knots, holds for braid positive ones, and for fibered positive knots is Stoimenow's conjecture, open, with a likely counterexample of genus 7 in [6].

6 Computations

All in `verification/families. unknotting.py`: for $2 \leq n \leq 12$ and $m \in \{1, 3, 5, 7\}$ (44 cases, up to 499 changes) the number of changes equals the slice–Bennequin bound and the changed word closes to the unknot (SnapPy simplification to no crossings, or knot Floer homology of total rank 1). `cables.py`: the two non-positive cases for $1 \leq k \leq 8$ (16 cables), with the genus from knot Floer homology. `fibered_sq.py`: the proposition. `lspace.py`: which $H(n, m)$ are L -space knots by knot Floer homology. Tools: SnapPy 3.3.2 with `knot_floer_homology`.

7 Questions

Question 1. *Is $u = g$ known, or expected, for every L -space knot?*

Question 2. *Besides the two families above and some iterated torus knots, which L -space knots are known not to be braid positive?*

References

- [1] K. L. Baker, M. Kegel, *Census L -space knots are braid positive, except for one that is not*, Algebr. Geom. Topol. 24 (2024); arXiv:2203.12013.
- [2] M. Hedden, *On knot Floer homology and cabling II*, Int. Math. Res. Not. (2009).
- [3] J. Hom, *A note on cabling and L -space surgeries*, Algebr. Geom. Topol. 11 (2011), 219–223.

- [4] M. Hedden, *Notions of positivity and the Ozsváth–Szabó concordance invariant*, J. Knot Theory Ramifications 19 (2010), 617–629.
- [5] K. Himeno, *Non-braid positive hyperbolic L -space knots*, arXiv:2506.22934.
- [6] M. Kegel, L. Lewark, N. Manikandan, F. Misev, L. Mousseau, M. Silvero, *On unknotting fibered positive knots and braids*, arXiv:2312.07339.
- [7] C. Livingston and A. H. Moore, *KnotInfo: Table of Knot Invariants*, knotinfo.org (accessed 14 September 2026).
- [8] E.-K. Lee, S.-J. Lee, *Unknotting number and genus of 3-braid knots*, J. Knot Theory Ramifications 22 (2013), 1350047; arXiv:1305.3455.
- [9] Y. Ni, *Knot Floer homology detects fibred knots*, Invent. Math. 170 (2007), 577–608.
- [10] Z. Nie, *An explicit description of $(1,1)$ L -space knots, and non-left-orderable surgeries*, arXiv:2102.10891.
- [11] P. Ozsváth, Z. Szabó, *On knot Floer homology and lens space surgeries*, Topology 44 (2005), 1281–1300.
- [12] L. Rudolph, *Braided surfaces and Seifert ribbons for closed braids*, Comment. Math. Helv. 58 (1983), 1–37.
- [13] L. Rudolph, *Quasipositivity as an obstruction to sliceness*, Bull. Amer. Math. Soc. 29 (1993), 51–59.
- [14] H. Wendt, *Die gordische Auflösung von Knoten*, Math. Z. 42 (1937), 680–696.